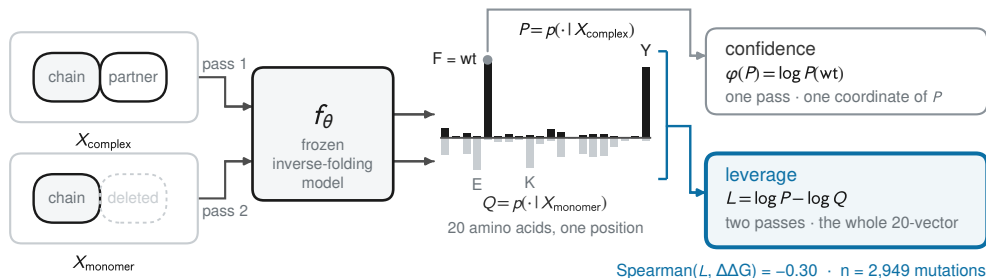


a confidence reads one coordinate of P; leverage reads the whole P → Q move

example position: 3SZK chain C F44 (a hotspot) · bars are the real 20-simplex



diffusion	$\varepsilon_\theta(x c)$	—	$\varepsilon_\theta(x \emptyset)$	classifier-free guidance is this same difference
this work	$\log p(\cdot X_{\text{complex}})$	—	$\log p(\cdot X_{\text{monomer}})$	

b confidence does not pin leverage down

13,401 positions · 343 complexes · 327 hotspots

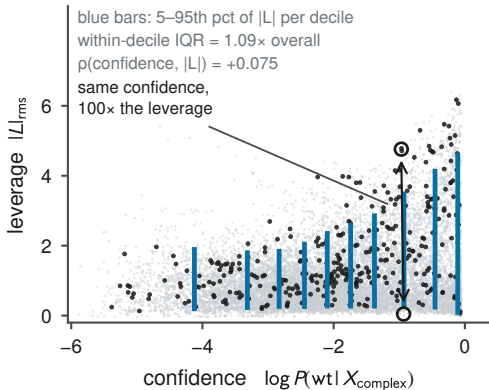
blue bars: 5–95th pct of $|L|$ per decile

within-decile IQR = 1.09× overall

$\rho(\text{confidence}, |L|) = +0.075$

same confidence,

100× the leverage

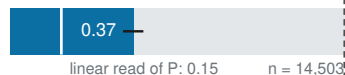


c

and neither does all of P

GBM / random forest, complex-held-out

ProteinMPNN 63% irreducible



ESM-IF1 64% irreducible



L unrelated to P

P would determine L

0.0 0.5 1.0

R^2 of $|L|_{\text{rms}}$ from all 20 coordinates of P